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Project Title:	Factors Contributing to Food Acceptability and Consumption, Mood, and Stress on Long-term Space Missions		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
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Space Biology Special Category:	None		
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Zip Code:	55108-1038	**Congressional District:**	4
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mann, Traci (University of Minnesota) Redden, Joseph (University of Minnesota)		
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Task Description:	Our goals for these studies were to examine the role of food and the eating experience as a strategy for lowering the stress or negative moods that astronauts might normally experience in flight, and to explore the variety of foods individuals consume and ways to increase their appreciation of the variety in their diet. To test whether preparing a meal and/or having the ability to choose the components of that meal would improve mood and reduce stress, we induced stress and overall negative mood in participants, and then measured whether that stress and negative mood were reduced by meal component choice or by meal preparation activities. People given no choice of meal components showed a greater decrease in negative moods and stress after eating a meal compared with those that were able to choose their meal components. To determine whether comfort foods repair individuals' moods, and if so, whether comfort foods were more effective at repairing moods than control foods or no food, participants came to a lab for separate sessions in which we induced a negative mood and then gave subjects a food (or no food) to eat. We measured their mood after the negative mood induction, and then again after eating (or not), to assess whether the food led to improvements in their mood. After watching films that induced a negative mood, participants' moods improved over time, but this happened to the same extent regardless of which type of food (comfort or equally liked non-comfort) they ate or whether they ate any food at all. In a separate study, participants were given chocolate either before or after we induced a negative mood, or were not given chocolate. In addition, before the film, some participants who were given chocolate were instructed to eat it right then, whereas others were instructed to save it for later. Participants who consumed chocolate before the film were significantly less upset by the film than participants not given chocolate and the participants that consumed chocolate after watching the film. Similarly, participants who simply received chocolate (but did not consume it) before the film were significantly less upset by the film than participants in the no food or the chocolate post-film conditions. Any effect of eating or receiving chocolate before the film disappeared by the final mood measure, however, in which no between group differences were found. To determine whether we could curb boredom with repeatedly eaten foods by increasing appreciation for variety in one's diet, participants repeatedly consumed granola bars over a 2-week period. Half the participants were reminded to think about recently eaten foods, and half were reminded to think about music they had recently heard. We were not able to prevent a decrease in granola bar liking (our measure of boredom) by reminding the participants to think about recently eaten foods (or music). To determine the amount of variety in free-living individuals, we had people keep 28-day diaries of all the foods they consumed. During that time participants ate an average of 109 unique food and beverage items. The average counts of unique items increased from breakfast (21) to lunch (38), and from lunch to dinner (46). Females had approximately 15% larger counts of unique items at meals than did males, and about 25% greater counts for snacks than did males.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of the role of food preparation and the eating situation on meal satisfaction, consumption, moods, and stress is highly related to these same issues on Earth. Is the selection and preparation of food 'therapy' for relieving stress? Or does it add to stress? And does this depend on specific characteristics of the individual? Information in the popular press strongly suggests that some foods (e.g., comfort foods) can relieve stress and repair bad moods. Does this really work on Earth? Are 'comfort foods' more effective at reducing stress and repairing moods than other foods? Than other activities? Our investigation of perceived variety is relevant for Earth-based situations where the variety of food people have available is limited or for people on highly restricted diets.
Task Progress:	To test the effects of meal preparation and control on moods and stress, we induced stress (and overall negative mood) in participants, and then measured whether that stress (and negative mood) was reduced by meal preparation activities. Post-meal responses indicated that stress and negative measures related to mood had decreased. Anxiety, anger, fatigue, and sadness, as well as systolic and diastolic blood pressure and cortisol decreased after the meal. Positivity rating increased after the meal. Heart rate did not change significantly from before to after the meal. Preparing a meal versus not preparing a meal had little effect on reducing stress or improving measures related to mood. Alternatively, heart rate increased more for those in the Prepare groups than in the No Prepare groups after the meal, and systolic blood pressure decreased more for those in the No Prepare groups versus the Prepare groups. Prepare versus No Prepare had no effect on the other responses. Those in the No Choice groups showed a greater decrease in anxiety, anger, and systolic blood pressure after the meal than those in the Choice groups. After the meal, heart rate increased more for those in the Choice versus the No Choice groups. Choice versus No Choice had no effect on the other responses. We conducted three versions of a study to assess whether comfort foods repair individuals' moods, and if so, whether comfort foods were more effective at repairing moods than control foods. Each of the three versions followed the same general design: participants first completed an online questionnaire about their food preferences (session one), then they came to the lab for two separate sessions in which we induced a negative mood and then gave subjects a food to eat. We measured their mood after the negative mood induction, and then again after they ate the food, to assess whether the food led to improvements in their mood. The most commonly provided comfort foods were chocolate (26%), ice cream (18%), cookies (11%), and brownies (8%). Eighty-one percent of participants either agreed or strongly agreed with the statement "I am confident that eating this food would make me feel better." The most commonly provided equally-liked non-comfort foods were almonds (15.1%), cashews (13.2%), brownies (9.4%), and popcorn (9.4%). Participants' negative moods improved over time, but this happened to the same extent regardless of which type of food they ate or whether they ate any food at all. In addition, it was not the case that participants received comfort only when they ate a food that they believed would comfort them, as there was no correlation between participants' confidence the food would comfort them and their change in mood. The amount of food they consumed also did not correlate with the change in mood. Consuming comfort food after the film was no more effective at improving mood than eating nothing after the film.

	There was some evidence that eating chocolate prior to the film, or even just receiving it prior to the film, may have reduced the negative effects of the film. Participants who consumed chocolate before the film were significantly less upset by the film than participants in the no food condition and the chocolate post-film condition. Similarly, participants who simply received chocolate before the film were significantly less upset by the film than participants in the no food or the chocolate post-film conditions. There was no difference in the moods of participants who consumed the chocolate or just received it to save for later. Any effect of eating or receiving chocolate before the film disappeared by the final mood measure, however, in which no between group differences were found. We asked participants to repeatedly consume granola bars over a 2-week period. Half the participants were reminded to think about recently eaten foods, and half were reminded to think about music they had recently heard. One hundred and forty-eight participants completed the study. As expected, participants displayed a significant decrease in liking of the granola bars over time. Mean liking of the granola bars was not greater in the experimental condition (reminded to think about recently eaten foods) than in control (reminded to think about musical artists they had listened to recently). However, a behavioral measure of willingness to continue eating granola bars did show that those in the experimental group were more willing to continue eating the bars. Participants in the experimental condition took significantly more bars home at the end of the study than did participants in the control condition. Participants in a 28-day food diary study ate on average 109 unique food and beverage items. For all participants, the average 28-day counts of unique items increased from breakfast to lunch and from lunch to dinner. Morning, afternoon, early evening, and late evening snacks showed little variation in count of unique items consumed. Female participants had approximately 15% larger counts of unique items than did male participants in their 28-day total and for breakfasts, lunches, and dinners. Females also displayed about 25% greater count of unique items for total snacks than did males.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Lenne RL, Mann T, Burns RJ, Vickers Z, Redden JP. "Variety salience and enjoyment of repetitiously consumed foods: A field experiment." Int J Behav Med. 2021 Jun;28(3):286-91. https://doi.org/10.1007/s12529-020-09916-2 ; PMID: 32601978 , Jun-2021